

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Orificed diagnostic fuel line, Part Number 3164621.

Additional Service Items

- No additional service items required.

General Information

Perform the steps in the following procedure before beginning this procedure to make sure all entrapped air is removed from the system. Refer to Procedure 005-234 in Section 5.
(</qs3/pubsys2/xml/en/procedures/377/377-005-234-tr.html>)

Setup

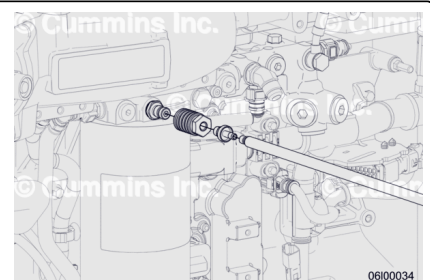
⚠ WARNING ⚠

High-pressure fuel spray can cause serious injury or death.

⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

⚠ WARNING ⚠



Do not bleed the fuel system of a hot engine; this can result in fuel spilling onto a hot exhaust manifold, which can cause a fire.

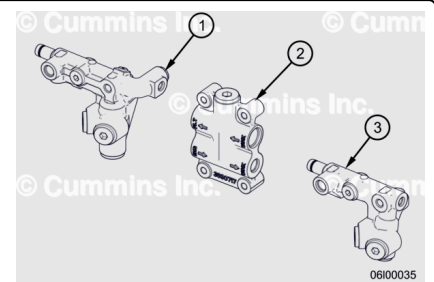
Install the orificed diagnostic fuel line, Part Number 3164621, onto the Compuchek™ fitting at the inlet to the pressure-side fuel filter. Place other end of orificed diagnostic fuel line in a bucket submerged under fuel. The orificed diagnostic fitting is used in procedures to create rated flow through the low-pressure fuel system without the need to operate the engine under load.

Measure

The following test method will simulate rated fuel flow through the system, so that air in fuel problems can be diagnosed.

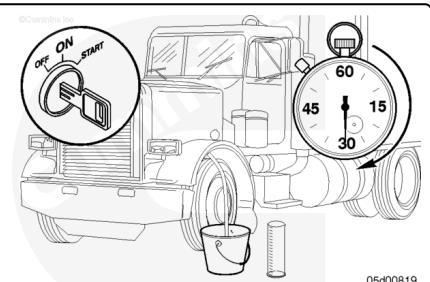
There are three variations of the Fuel Thermal Recirculation Valve.

1. Drain manifold integrated thermal recirculation valve.
2. Upfit thermal recirculation valve.
3. Drain manifold integrated thermal recirculation valve with check valve.



Note : A symptom of air in fuel for the engines equipped with a common rail fuel system is an audible surge associated with fuel system pressure fluctuations, due to air in the fuel supply. Air bleed time is dependent on fuel temperature. Units with a thermal recirculation valve (TRV) will experience a longer air bleed time when the fuel temperature is at or below 25°C [77°F]. Three fuel lift pump cycles will allow for an accurate assessment of air in fuel.

Start the engine and operate the engine speed from idle to high idle several times to purge the air induced when the diagnostic fuel line was installed.



Observe the fuel flow exiting the diagnostic fuel line while the engine operates at idle speed.

A few bubbles exiting the line is expected. A foamy appearance is an indication of a leak that allows air to enter, a severe inlet restriction that causes cavitation, or a system that is **not** yet primed.

If the engine will **not** start, cycle the fuel lift pump three times, each time observing for excessive air bubbles exiting the orificed diagnostic fuel line.

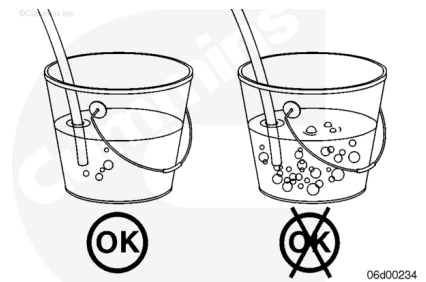
Use the following procedure to check inlet restriction. Refer to Procedure 006-020 in Section 6.

(/qs3/pubsys2/xml/en/procedures/377/377-006-020.html) If the fuel inlet restriction is **not** excessive, the source of air entry **must** be isolated to one of the following:

- Suction fuel lines
- Original equipment manufacturer (OEM) fuel lines
- Suction-side fuel filter assemblies
- Lift pump gasket or fittings
- Stand pipe(s) in the fuel tank(s)

Use the following procedure if equipped with a drain manifold fuel thermal recirculation valve. Refer to Procedure 006-021 in Section 6. (/qs3/pubsys2/xml/en/procedures/377/377-006-021.html)

See equipment manufacturer service information if equipped with an up-fit fuel thermal recirculation valve.

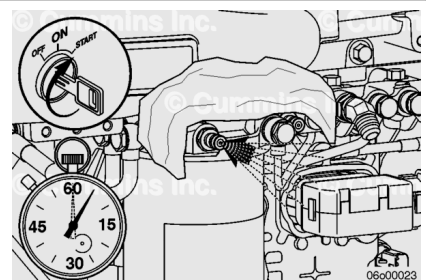


Test

If the engine is hard to start or will **not** start, perform the following test to check for trapped air.

Note : Prior to depressing the Compuchek™ fitting, cover the fitting with a shop towel to contain any possible fuel leakage.

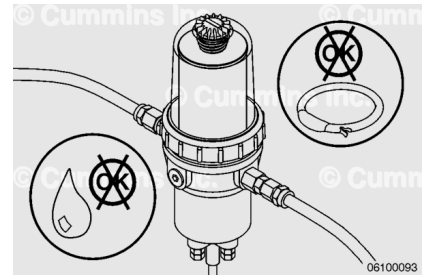
- Crank the engine for five seconds (failed starting attempt).
- Immediately after the failed starting attempt, depress the center of the Compuchek™ fitting on the pressure filter.
- A solid stream of fuel indicates no air is present in the system.



- A spray of fuel during this test confirms there is trapped air in the system.

If air is found to be trapped in the system, it **must** be primed. If the system can **not** maintain prime, there could be suction-side fuel leaks permitting continual air entry.

If air is found to be trapped in the system, inspect the OEM supplied suction-side fuel filter for leaks, loose fittings, damaged o-rings, etc.



Finishing Steps

- Remove all test fuel lines.
- Operate the engine. Check for leaks.